Decorrelated Debiased Converted Measurement Kalman Filter for Phased Array Radar Tracking System

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In this article, the target tracking of phased array radars with measurements specified in the direction cosine coordinate system is considered. The converted measurement Kalman filter is a widely adopted technique since it allows tracking in the coordinate system in which the dynamics of targets are described. The converted measurement Kalman filter has two types of inherent bias: conversion and estimation biases. The compensation of conversion bias has been examined by earlier studies using the debiasing technique in the presence of estimation bias. This article, therefore, introduces decorrelated debiased converted measurement Kalman filter (DDCMKF) in the direction cosine coordinate system to relieve the estimation bias. The decorrelation between the Kalman gain and measurement residual by applying conditioning on predicted position estimates is first introduced. The range-rate measurement is subsequently incorporated as a pseudo-measurement form after the derivation of decorrelated converted position. Finally, a sequential DDCMKF in direction cosine coordinates is developed using the derived converted position and range-rate measurement models. Adopting sequential filtering for incorporating the range-rate measurement reduces computation complexity and alleviates its strong nonlinearity. Numerical simulations are conducted to evaluate the performance of DDCMKF against other types of tracking filters in the direction cosine coordinate system. Simulation results show that the proposed DDCMKF achieves preferable error performance while maintaining consistency, especially in short-range tracking with relatively large direction cosine errors.

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I. INTRODUCTION

A Phased array radar is developed to track faster, more accurately with wide scan coverage. The phased array supports a track-while-scan mode and shows improved situational awareness compared with conventional mechanically steered arrays. The phased array radar yields its measurements in a direction cosine coordinate system rather than a spherical coordinate system because directing a beam can be easily controlled by using two-direction cosines to radar plane axes [1], [2]. Since a measurement specified in direction cosine coordinates is a nonlinear function of target states in a radar Cartesian coordinate system (RCCS), tracking the target in radar coordinates turns into a nonlinear state estimation problem.

The converted measurement Kalman filter (CMKF) is a broadly adopted method for a nonlinear target tracking problem. Measurements in sensor coordinates are nonlinearly transformed to the state in the tracking coordinate system, and then a linear Kalman filter is applied in the CMKF. The linear CMKF gives good accuracy with reduced computational burden if a consistent transformation of measurements is accompanied. Tracking with the CMKF in the spherical coordinate system has been extensively studied [3]–[10]. The CMKF possesses two potential bias issues caused by the nonlinear transformation in it. One is a conversion bias that originates from the nonlinear transformation of measurements. The other is an estimation bias that arises out of a correlation between the measurement error covariance and the measurement noise. The conversion bias can be compensated through a debiased conversion [3], [4] or an unbiased conversion [5]–[7]. The estimation bias is alleviated through the measurement error covariance conditioning on the predicted estimate [8], [9] or the previous measurement [10]. Contrary to many study cases about the CMKF in the spherical coordinate system, fewer studies about the CMKF in the direction cosine coordinate system exist [2], [11]–[13]. As noted in [13], a closed-form unbiased measurement conversion using direction cosine measurements does not exist. Previous studies, therefore, have focused on deriving the consistent mean and the covariance of conversion error, which is used in the debiased CMKF in the direction cosine coordinate system [2], [11], [12]. Though the conversion bias originating from direction cosine measurements can be alleviated using debiasing, the estimation bias still stems from the correlation between the measurement-dependent covariance and the measurement noise. Further techniques for reducing the estimation bias need to be developed to improve the performance of CMKF in direction cosine coordinates, which leads to a rudimentary idea, an application of an approximated decorrelation in our previous study [20].

Most recent electronically scanned array radar systems provide range-rate measurements as well as three-dimensional (3-D) position measurements. The range-rate measurement increases the observability of target states. However, range-rate measurements exhibit significant nonlinearity with the correlation between range measurements and those under some waveforms [14]. Commonly used nonlinear filtering techniques such as the extended Kalman filter (EKF) [15], the linearized Kalman filter [19], and the unscented Kalman filter (UKF) are applicable. Nonetheless, those techniques may yield degraded performance due to the strong nonlinearity of range-rate measurements. The sequentialization of the measurement update as the EKF in [16] and [17], and the linearized Kalman filter in [18] were also proposed to improve the accuracy of the range-rate measurement update using estimated states from the position update. The augmentation of a second-order EKF to the conventional position CMKF (DCMSEKF) introduced in [16] and [17] easily incorporates range-rate measurements in the form of pseudo-measurements. However, the DCMKSEKF may be susceptible to measurement noise level since the inaccuracy of the position estimation affects the recursive EKF unfavorably. The DCMSEKF, moreover, requires complex computation for the second-order filtering. The UKF can consider range-rate measurements [21], [22] as well. Still, it requires more calculation for the unscented transformation and may encounter a consistency problem, especially in tracking distant targets. An algorithm using noninformative cross-range-rates components [23], [24] was proposed to avoid estimation accuracy degradation due to nonlinearity, which adopts an information-form Kalman filter. However, methods become sensitive to the initial standard deviation of target velocity. The static fusion of the position CMKF (PCMKF) and the range-rate CMKF (DCMKF) suggested in [13] showed preferable performance under highly nonlinear tracking conditions. Still, the application to general dynamic models might be limited since it requires the derivation of appropriate pseudo-state equations for the specific dynamic model used in the filter.

Inspired by previous considerations, this article develops a decorrelated debiased converted measurement Kalman filter (DDCMKF) in direction cosine coordinates with range-rate measurement incorporated. A thorough derivation and a demonstration of its performance are mainly discussed. First, the position conversion in direction cosine coordinates considering a bias compensation and a decorrelation is developed. The decorrelation using predicted estimates in direction cosine coordinates is introduced. Second, the pseudo-linearization of range-rate measurement using estimates from the prior update is explained and incorporated in the measurement update. The corresponding decorrelated bias and error covariance of pseudo-linearized range-rate measurement are also acquired. The sequential filtering of the pseudo-linearized range rate, in addition, is suggested to reduce computation complexity while fully utilizing position estimation results. Finally, two illustrative scenarios of Monte-Carlo simulations demonstrate the performance of the proposed DDCMKF, which simultaneously compare the error and the consistency with other nonlinear filters.

The rest of this article is organized as follows. In Section II, the target tracking problem with position and range-rate measurements in direction cosine coordinates is defined. Section III shows a derivation of the bias and error covariance of converted position measurements and pseudo-linearized range-rate measurements. The algorithm of the DDCMKF and how different it is from other nonlinear filter algorithms are explained subsequently. Section IV presents the results of consistency tests and Monte-Carlo simulation results to assesses the proposed DDCMKF compared with existing filters with direction cosine measurements. Finally, Section V concludes this article.

II. PROBLEM FORMULATION

A. Target Dynamic Model

The target dynamics is usually modeled as a linear discrete difference equation in the RCCS as follows:

$$\mathbf{x}_{k+1} = \mathbf{F}_k \mathbf{x}_k + \mathbf{w}_k. \tag{1}$$

where $\mathbf{x}_k$ represents the state of a target, $\mathbf{F}_k$ is a state transition matrix, and $\mathbf{w}_k$ is a zero-mean Gaussian process noise with covariance $\mathbf{Q}_k$ at the filtering step k. The target state vector $\mathbf{x}_k$ in (1) consists of a 3-D position and velocity components in Cartesian coordinates as follows:

$$\mathbf{x}_k = [x_k, y_k, z_k, \dot{x}_k, \dot{y}_k, \dot{z}_k]^T. \tag{2}$$

Note that the process noise is given in a linear additive form. Among many discrete-time target models introduced in [25], the constant velocity model with small white noise acceleration is considered. The small white noise acceleration illustrates small perturbations from the environment like turbulence or gust. The constant velocity model is specified as follows:

$$\mathbf{F}_k = \begin{bmatrix} \mathbf{I}_{3\times 3} & T\,\mathbf{I}_{3\times 3} \\ \mathbf{0}_{3\times 3} & \mathbf{I}_{3\times 3} \end{bmatrix}$$

$$\mathbf{Q}_k = E\left[\mathbf{w}_k \mathbf{w}_k^T\right] = \begin{bmatrix} \frac{T^3}{3}\mathbf{q} & \frac{T^2}{2}\mathbf{q} \\ \frac{T^2}{2}\mathbf{q} & T\mathbf{q} \end{bmatrix}, \quad \mathbf{q} = \begin{bmatrix} q_x & 0 & 0 \\ 0 & q_y & 0 \\ 0 & 0 & q_z \end{bmatrix} \tag{3}$$

where the components of process noise vector $\mathbf{w}_k$ for each axis are assumed to be uncorrelated with other axes components, and their power spectral densities are denoted as q_x, q_y, and q_z. The sampling interval for the discrete model is denoted as T.

B. Measurement Model

The measurements for the phased array radar tracking system are given in the sensor coordinate system, direction cosine coordinates as shown in Fig. 1. The phased array radar provides target information in a range, a range rate and direction cosines of a line-of-sight vector with respect to array axes, x and y. The measurement equation is given as follows:

$$\mathbf{z}_k = h(\mathbf{x}_k) + \mathbf{v}_k = \left[\, r_{m,k} \;\; \alpha_{m,k} \;\; \beta_{m,k} \;\; \dot{r}_{m,k} \,\right]^T$$

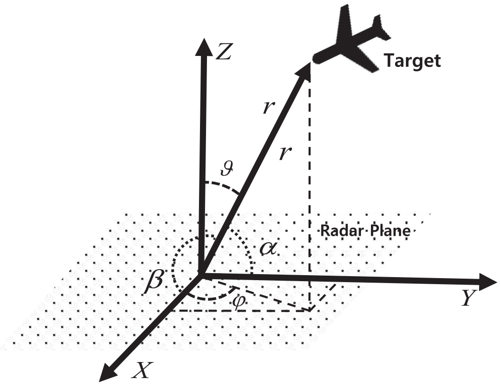


Fig. 1. Measurements in the direction cosine coordinate system.

$$= \begin{bmatrix} \sqrt{x_k^2 + y_k^2 + z_k^2} \\ x_k / \sqrt{x_k^2 + y_k^2 + z_k^2} \\ y_k / \sqrt{x_k^2 + y_k^2 + z_k^2} \\ (x_k \dot{x}_k + y_k \dot{y}_k + z_k \dot{z}_k) / \sqrt{x_k^2 + y_k^2 + z_k^2} \end{bmatrix} + \begin{bmatrix} v_{r,k} \\ v_{\alpha,k} \\ v_{\beta,k} \\ v_{\dot{r},k} \end{bmatrix} \tag{4}$$

where a measurement vector $\mathbf{z}_k$ at the filtering step k consists of $r_{m,k}$, $\alpha_{m,k}$, $\beta_{m,k}$, and $\dot{r}_{m,k}$. Each component denotes a range, x- and y-axis direction cosines and a range rate. A noise vector $\mathbf{v}_k$ contains $v_{r,k}$, $v_{\alpha,k}$, $v_{\beta,k}$, and $v_{\dot{r},k}$ that are measurement noises of each element. Noise components are assumed to be zero-mean Gaussian with known variances σ_r^2, σ_α^2, σ_β^2, and $\sigma_{\dot{r}}^2$, respectively. They are supposed to be mutually independent except that range and range-rate noise, $v_{r,k}$ and $v_{\dot{r},k}$, are correlated with coefficient ρ. The covariance matrix of the measurements in the direction cosine coordinates are given as follows:

$$\mathbf{R}_k = \begin{bmatrix} \sigma_r^2 & 0 & 0 & \rho\sigma_r\sigma_{\dot{r}} \\ 0 & \sigma_\alpha^2 & 0 & 0 \\ 0 & 0 & \sigma_\beta^2 & 0 \\ \rho\sigma_r\sigma_{\dot{r}} & 0 & 0 & \sigma_{\dot{r}}^2 \end{bmatrix}. \tag{5}$$

Fig. 1 illustrates the off-boresight angle ϑ as an angle between the plane normal z-axis and line-of-sight. The off-boresight angle is more broadly adopted in describing antenna geometry in radar systems than a polar angle in spherical coordinates. The azimuth angle φ is a horizontal angle between the x-axis and the projected line-of-sight on the radar plane. The nonlinear conversion of measurement enables applying a linear recursive estimator in spite of measurements expressed in nonlinear equations of radar Cartesian coordinates.

III. DECORRELATED DEBIASED CMKF WITH RANGE-RATE MEASUREMENT

The derivation of the DDCMKF incorporating the range-rate measurement is provided in this section. The derivation of the corresponding moments of converted measurement error statistics, the inclusion of the pseudo-linearized range rate using the sequential filtering, and details of the DDCMKF algorithm are explained.

A. Decorrelated Debiased Position Conversion

The measurement conversion aims to express nonlinear measurements in terms of linear functions of the states in radar Cartesian coordinates [5], [6], [8]. In the direction cosine coordinate system, the position conversion from the raw measurement (4) to the position in radar Cartesian coordinates is achieved as follows:

$$\mathbf{z}_{pos} = \begin{bmatrix} x_c \\ y_c \\ z_c \end{bmatrix} = \begin{bmatrix} r_m\alpha_m \\ r_m\beta_m \\ r_m\gamma_m \end{bmatrix} = \begin{bmatrix} r_m\alpha_m \\ r_m\beta_m \\ r_m\sqrt{1-\alpha_m^2-\beta_m^2} \end{bmatrix}. \tag{6}$$

Note that the subscript k denoting a filtering step might be omitted in the following equations where they are unnecessary. Since the z-axis direction cosine γ is not directly measured, the z-axis coordinate z_m is expressed in terms of the nonlinear function of the other direction cosine measurements. No closed-form unbiased conversion can be found for the z-axis direction cosine γ_m. The third direction cosine error v_γ can be only expressed using Taylor expansion of fourth-order as follows:

$$\begin{aligned} v_\gamma &= \gamma_\alpha v_\alpha + \gamma_\beta v_\beta \\ &\quad + \left(\frac{1}{2}\gamma_{\alpha\alpha} v_\alpha^2 + \gamma_{\alpha\beta} v_\alpha v_\beta + \frac{1}{2}\gamma_{\beta\beta} v_\beta^2\right) \\ &\quad + \left(\frac{1}{6}\gamma_{\alpha\alpha\alpha} v_\alpha^3 + \frac{1}{2}\gamma_{\alpha\alpha\beta} v_\alpha^2 v_\beta + \frac{1}{2}\gamma_{\alpha\beta\beta} v_\alpha v_\beta^2 \right. \\ &\quad \left. + \frac{1}{6}\gamma_{\beta\beta\beta} v_\beta^3\right) \\ &\quad + \left(\frac{1}{24}\gamma_{\alpha\alpha\alpha\alpha} v_\alpha^4 + \frac{1}{6}\gamma_{\alpha\alpha\alpha\beta} v_\alpha^3 v_\beta + \frac{1}{4}\gamma_{a\alpha\beta\beta} v_\alpha^2 v_\beta^2 \right. \\ &\quad \left. + \frac{1}{6}\gamma_{a\beta\beta\beta} v_\alpha v_\beta^3 + \frac{1}{24}\gamma_{\beta\beta\beta\beta} v_\beta^4\right) + R_5(v_\alpha, v_\beta) \end{aligned} \tag{7}$$

where higher-order terms represented as R_5 are neglected in the following formulations. Equations of partial derivatives of the third direction cosine γ from the first derivative γ_α to the fourth derivative $\gamma_{\beta\beta\beta\beta}$ are given in (35) of Appendix A. Although the second-order Taylor expansion to approximate the direction cosine error v_γ used in [2] and [13] yields a precise error estimate to a practical extent, the fourth-order approximation of v_γ as in [12] is applied to estimate the first two moments of converted position errors accurately. Employing a higher-order approximation further extends a geometric tracking range.

From (4), the converted position error is given as follows:

$$\tilde{\mathbf{z}}_{pos} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{bmatrix} = \begin{bmatrix} x_c - x \\ y_c - y \\ z_c - z \end{bmatrix} = \begin{bmatrix} r_t v_\alpha + \alpha_t v_r + v_r v_\alpha \\ r_t v_\beta + \beta_t v_r + v_r v_\beta \\ r_t v_\gamma + \gamma_t v_r + v_r v_\gamma \end{bmatrix} \tag{8}$$

where $\tilde{x}$, $\tilde{y}$, and $\tilde{z}$ are the converted position errors, and r_t, α_t, and β_t denote true range and direction cosines. Since true states r_t, α_t, and β_t are unavailable in practical tracking, the converted measurement error in (8) should be conditioned on known values to compute the error bias and covariance of the converted measurement. Conditioning on measurements proposed in [2], [12], [13], and [17] would be a possible resolution to substitute unavailable true states. However, the acquired error covariance and bias expressed in terms of measurements would create the correlation between Kalman gain and measurement residual, which leads to significant bias in the estimation. Besides, the measurement usually contains more significant errors than the filter's estimation. Using predicted estimates on conditioning the bias and covariance [8], [9] or on approximating covariance [26], [27] has been proposed to solve these limitations. The decorrelation scheme using the predicted estimate that was proved to be efficient in spherical coordinates [8], [9] is adopted. To calculate the decorrelated error covariance and bias, rewrite (8) as follows:

$$\tilde{\mathbf{z}}_{pos} = \begin{bmatrix} \tilde{x} \\ \tilde{y} \\ \tilde{z} \end{bmatrix} = \begin{bmatrix} (r_p - \lambda_r)\upsilon_\alpha + (\alpha_p - \lambda_\alpha)\upsilon_r + \upsilon_r\upsilon_\alpha \\ (r_p - \lambda_r)\upsilon_\beta + (\beta_p - \lambda_\beta)\upsilon_r + \upsilon_r\upsilon_\beta \\ (r_p - \lambda_r)\upsilon_\gamma + (\gamma_p - \lambda_\gamma)\upsilon_r + \upsilon_r\upsilon_\gamma \end{bmatrix}. \tag{9}$$

In (9), estimated states in direction cosine coordinates r_p, α_p, β_p, and γ_p replace unknown true states. The estimation errors of range, three-direction cosines are denoted as λ_r, λ_α, λ_β, and λ_γ each. They are assumed to be zero-mean, Gaussian with covariance $\mathbf{\Lambda}_{pos}$. Defining the predicted estimate vector as $\hat{\mathbf{z}}_{pos} = [r_p, \alpha_p, \beta_p, \gamma_p]^T$, the estimated states and their covariance in direction cosine coordinates are computed as (10) from the *a priori* estimate $\hat{\mathbf{x}}_p = [x_p, y_p, z_p, \dot{x}_p, \dot{y}_p, \dot{z}_p]^T$ and its error covariance estimate $\mathbf{P}_p$

$$\begin{aligned} \hat{\mathbf{z}}_p &= h_1(\hat{\mathbf{x}}_p) = \left[\, r_p \; \alpha_p \; \beta_p \; \gamma_p \,\right]^T \\ &= \left[\sqrt{x_p^2 + y_p^2 + z_p^2} \; x_p r_p^{-1} \; y_p r_p^{-1} \; z_p r_p^{-1} \right]^T \\ \mathbf{\Lambda}_{pos} &= \mathbf{J}_1 \mathbf{P}_p \mathbf{J}_1^T = \begin{bmatrix} \tau_{rr} & \tau_{r\alpha} & \tau_{r\beta} & \tau_{r\gamma} \\ \tau_{r\alpha} & \tau_{\alpha\alpha} & \tau_{\alpha\beta} & \tau_{\alpha\gamma} \\ \tau_{r\beta} & \tau_{\alpha\beta} & \tau_{\beta\beta} & \tau_{\beta\gamma} \\ \tau_{r\gamma} & \tau_{\alpha\gamma} & \tau_{\beta\gamma} & \tau_{\gamma\gamma} \end{bmatrix} \end{aligned} \tag{10}$$

where the covariance in direction cosine coordinates is approximated by linearly transforming the error covariance in Cartesian coordinates. The Jacobian matrix $\mathbf{J}_1$ is

$$\begin{aligned} \mathbf{J}_1 &= \left.\frac{\partial \mathbf{h}_1}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_p} = \left[\left.\tfrac{\partial r}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_p} \; \left.\tfrac{\partial \alpha}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_p} \; \left.\tfrac{\partial \beta}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_p} \; \left.\tfrac{\partial \gamma}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_p} \right]^T \\ \frac{\partial r}{\partial \mathbf{x}} &= r^{-1}\left[\, x \; y \; z \; \mathbf{0}_{1\times 3} \,\right]^T \\ \frac{\partial \alpha}{\partial \mathbf{x}} &= r^{-3}\left[\, y^2 + z^2 \; -xy \; -xz \; \mathbf{0}_{1\times 3} \,\right]^T \\ \frac{\partial \beta}{\partial \mathbf{x}} &= r^{-3}\left[\, -xy \; x^2 + z^2 \; -yz \; \mathbf{0}_{1\times 3} \,\right]^T \\ \frac{\partial \gamma}{\partial \mathbf{x}} &= r^{-3}\left[\, -xz \; -yz \; x^2 + y^2 \; \mathbf{0}_{1\times 3} \,\right]^T. \end{aligned} \tag{11}$$

Then, the bias and error covariance are obtained from (9) as follows:

$$\mu_{pos} = E\left[\tilde{\mathbf{z}}_{pos} \middle| \hat{\mathbf{z}}_{pos} \right] = \left[\, 0 \; 0 \; r_p E\left[\upsilon_\gamma \middle| \alpha_p, \beta_p \right] \right]^T \tag{12}$$

$$\begin{aligned} \mathbf{R}_{pos} &= E\left[\left(\tilde{\mathbf{z}}_{pos} - \mu_{pos}\right)\left(\tilde{\mathbf{z}}_{pos} - \mu_{pos}\right)^T \middle| \hat{\mathbf{z}}_{pos} \right] \\ &= \begin{bmatrix} R_{xx} & R_{xy} & R_{xz} \\ R_{xy} & R_{yy} & R_{yz} \\ R_{xz} & R_{yz} & R_{zz} \end{bmatrix} \end{aligned} \tag{13}$$

$$\begin{aligned} R_{xx} &= (r_p^2 + \tau_{rr})\sigma_\alpha^2 + (\alpha_p^2 + \tau_{\alpha\alpha})\sigma_r^2 + \sigma_r^2\sigma_\alpha^2 \\ R_{yy} &= (r_p^2 + \tau_{rr})\sigma_\beta^2 + (\beta_p^2 + \tau_{\beta\beta})\sigma_r^2 + \sigma_r^2\sigma_\beta^2 \\ R_{zz} &= (r_p^2 + \tau_{rr} + \sigma_r^2)E\left[\upsilon_\gamma^2 \middle| \alpha_p, \beta_p \right] + (\gamma_p^2 + \tau_{\gamma\gamma})\sigma_r^2 \\ &\quad + 2\gamma_p\sigma_r^2 E\left[\upsilon_\gamma \middle| \alpha_p, \beta_p \right] - r_p^2 E^2\left[\upsilon_\gamma \middle| \alpha_p, \beta_p \right] \\ R_{xy} &= (\alpha_p\beta_p + \tau_{\alpha\beta})\sigma_r^2 \\ R_{xz} &= (r_p^2 + \tau_{rr} + \sigma_r^2)E\left[\upsilon_\alpha\upsilon_\gamma \middle| \alpha_p, \beta_p \right] \\ &\quad + (\alpha_p\gamma_p + \tau_{\alpha\gamma})\sigma_r^2 + \alpha_p\sigma_r^2 E\left[\upsilon_\gamma \middle| \alpha_p, \beta_p \right] \\ R_{yz} &= (r_p^2 + \tau_{rr} + \sigma_r^2)E\left[\upsilon_\beta\upsilon_\gamma \middle| \alpha_p, \beta_p \right] \\ &\quad + (\beta_p\gamma_p + \tau_{\beta\gamma})\sigma_r^2 + \beta_p\sigma_r^2 E\left[\upsilon_\gamma \middle| \alpha_p, \beta_p \right] \end{aligned} \tag{14}$$

where expectations $E[\upsilon_\gamma|\alpha_p, \beta_p]$, $E[\upsilon_\alpha\upsilon_\gamma|\alpha_p, \beta_p]$, $E[\upsilon_\beta\upsilon_\gamma|\alpha_p, \beta_p]$, and $E[\upsilon_\gamma^2|\alpha_p, \beta_p]$ in (12) and (14) are calculated from conditioning the third-direction cosine error with the predicted estimate. Appendix A introduces how to approximate and calculate these expectations.

Remark that the decorrelation by substituting predicted estimates instead of measurements prevent the algorithm from showing the estimation bias caused by the correlated measurement residual and the error covariance during the update step. The error covariance and the bias expressed in terms of the filter's prediction possess different forms to those using measurements as in [2], [12], and [17]. Equations (10) and (11) are obtained based on the assumption that the estimated error statistics are Gaussian in the direction cosine coordinate system. The Gaussianity assumption of the error statistics can be employed in both direction cosine and Cartesian coordinates. The innovation covariance of the recursive linear minimum mean-squared error (LMMSE) filter in [26] and [27] is an example derived by supposing Gaussianity in Cartesian coordinates. The proposed decorrelated conversion could provide superior initial convergence performance. It is because the true statistics of estimated errors remain unknown, and a Gaussianity assumption in direction cosine coordinates would be more valid at the early steps of the tracking. The proposed conversion is also computationally light in that it does not require the unscented transformation for the covariance estimation as in [27].

B. Pseudo-Linearization of Range-Rate Measurement

Using the range-rate measurement draws out more information about the target state. It is a highly nonlinear function and cannot be directly converted to velocity components in Cartesian coordinates without angular rate measurements. The pseudo-linearization of it is suggested to cope with the nonlinearity and to keep the linear Kalman filter structure. From (4), the range rate can be rewritten in linearized form using true direction cosines $(\alpha_t, \beta_t, \gamma_t)$ and true velocity in radar Cartesian coordinates as follows:

$$\dot{r}_m = \alpha_t \dot{x}_t + \beta_t \dot{y}_t + \gamma_t \dot{z}_t + v_{\dot{r}} \tag{15}$$

Then, the linearized radial velocity in (15) can be embodied directly in the converted measurement model. Still, two problems exist: the correlation between the range rate and the range remains, and the true direction cosines used for linearization in (15) are unavailable in an actual implementation. The Cholesky decomposition removes the correlation with the range measurement error from the range-rate measurement noise $v_{\dot{r}}$. Adopting it also facilitates the application of the sequential filtering method in the DDCMKF algorithm. Apply the decomposition to the part of the measurement error covariance, then

$$\mathbf{R}' = \begin{bmatrix} \sigma_r^2 & \rho\sigma_r\sigma_{\dot{r}} \\ \rho\sigma_r\sigma_{\dot{r}} & \sigma_{\dot{r}}^2 \end{bmatrix} = \mathbf{L}\begin{bmatrix} \sigma_r^2 & 0 \\ 0 & (1-\rho^2)\sigma_{\dot{r}}^2 \end{bmatrix}\mathbf{L}^T$$
$$\mathbf{L} = \begin{bmatrix} 1 & 0 \\ \rho\sigma_{\dot{r}}\sigma_r^{-1} & 1 \end{bmatrix}. \tag{16}$$

Repeat the decomposition to the range and range-rate equation, then

$$\begin{bmatrix} r_m \\ \dot{r}_m - \rho\sigma_{\dot{r}}\sigma_r^{-1} r_m \end{bmatrix} = \begin{bmatrix} \mathbf{\Delta}_t & \mathbf{0}_{1\times 3} \\ -\rho\sigma_{\dot{r}}\sigma_r^{-1}\mathbf{\Delta}_t & \mathbf{\Delta}_t \end{bmatrix}\mathbf{x} + \begin{bmatrix} v_r \\ v_{\dot{r}} - \rho\sigma_{\dot{r}}\sigma_r^{-1} v_r \end{bmatrix} \tag{17}$$

where $\mathbf{\Delta}_t = [\alpha_t,\ \beta_t,\ \gamma_t]$ represents the true direction cosine and $\mathbf{x}$ is the true state vector as in (2). The linearized range rate in (15) now turns into a new pseudo-measurement that is mutually independent of the range measurement. Note that the range equation is unaffected by the decomposition, whereas the pseudo-radial velocity component is formulated to the linear combination of two measurements. Then, the true direction cosine in (17), which is unavailable in an actual implementation, is substituted with the estimated direction cosine

$$\begin{aligned} \mathbf{z}_{rr} &= \dot{r}_m - \rho\sigma_{\dot{r}}\sigma_r^{-1} r_m \\ &= -\rho\sigma_{\dot{r}}\sigma_r^{-1}(\alpha_p x_t + \beta_p y_t + \gamma_p z_t) \\ &\quad + (\alpha_p \dot{x}_t + \beta_p \dot{y}_t + \gamma_p \dot{z}_t) + \eta_{\dot{r}}, \end{aligned} \tag{18}$$

where the noise of the pseudo-linearized range-rate measurement $\eta_{\dot{r}}$ is given as follows:

$$\begin{aligned} \eta_{\dot{r}} &= (v_{\dot{r}} - \rho\sigma_{\dot{r}}\sigma_r^{-1} v_r) \\ &\quad + \rho\sigma_{\dot{r}}\sigma_r^{-1}(\lambda_\alpha x_t + \lambda_\beta y_t + \lambda_\gamma z_t) \\ &\quad - (\lambda_\alpha \dot{x}_t + \lambda_\beta \dot{y}_t + \lambda_\gamma \dot{z}_t). \end{aligned} \tag{19}$$

In analogy to the decorrelation of converted position, true states x_t, y_t, z_t, $\dot{x}_t$, $\dot{y}_t$, and $\dot{z}_t$ in the noise $\eta_{\dot{r}}$ are now expressed via estimated states and estimation errors in direction cosine coordinates to get

$$\begin{aligned} \eta_{\dot{r}} &= (v_{\dot{r}} - \rho\sigma_{\dot{r}}\sigma_r^{-1} v_r) \\ &\quad + \rho\sigma_{\dot{r}}\sigma_r^{-1}\lambda_\alpha(r_p - \lambda_r)(\alpha_p - \lambda_\alpha) \\ &\quad - \lambda_\alpha\{(\dot{r}_p - \lambda_{\dot{r}})(\alpha_p - \lambda_\alpha) + (r_p - \lambda_r)(\dot{\alpha}_p - \lambda_{\dot{\alpha}})\} \\ &\quad + \rho\sigma_{\dot{r}}\sigma_r^{-1}\lambda_\beta(r_p - \lambda_r)(\beta_p - \lambda_\beta) \\ &\quad - \lambda_\beta\{(\dot{r}_p - \lambda_{\dot{r}})(\beta_p - \lambda_\beta) + (r_p - \lambda_r)(\dot{\beta}_p - \lambda_{\dot{\beta}})\} \\ &\quad + \rho\sigma_{\dot{r}}\sigma_r^{-1}\lambda_\gamma(r_p - \lambda_r)(\gamma_p - \lambda_\gamma) \\ &\quad - \lambda_\gamma\{(\dot{r}_p - \lambda_{\dot{r}})(\gamma_p - \lambda_\gamma) + (r_p - \lambda_r)(\dot{\gamma}_p - \lambda_{\dot{\gamma}})\}. \end{aligned} \tag{20}$$

where estimated states $r_p, \alpha_p, \beta_p, \gamma_p$ and their corresponding error components $\lambda_r, \lambda_\alpha, \lambda_\beta, \lambda_\gamma$ are identical to states given in (10). The other estimated velocity states $\dot{r}_p$, $\dot{\alpha}_p$, $\dot{\beta}_p$, $\dot{\gamma}_p$ and their corresponding errors $\lambda_{\dot{r}}$, $\lambda_{\dot{\alpha}}$, $\lambda_{\dot{\beta}}$, $\lambda_{\dot{\gamma}}$ are further assumed to be zero-mean Gaussian as same as position estimation errors. Then, the predicted estimate vector $\hat{\mathbf{z}}_{rr} = [r_p,\ \alpha_p,\ \beta_p,\ \gamma_p,\ \dot{r}_p,\ \dot{\alpha}_p,\ \dot{\beta}_p,\ \dot{\gamma}_p]^T$ and their covariance $\mathbf{\Lambda}_{rr}$ in direction cosine coordinates are computed from the *a priori* estimate $\hat{\mathbf{x}}_p$ and its error covariance $\mathbf{P}_p$ as in (21) similar to position conversion

$$\begin{aligned} \hat{\mathbf{z}}_{rr} &= [h_1(\hat{\mathbf{x}}_p)\ h_2(\hat{\mathbf{x}}_p)]^T \\ &= \left[r_p\ \alpha_p\ \beta_p\ \gamma_p\ \dot{r}_p\ \dot{\alpha}_p\ \dot{\beta}_p\ \dot{\gamma}_p \right]^T \\ &= \begin{bmatrix} h_1(\hat{\mathbf{x}}_p) \\ (x_p\dot{x}_p + y_p\dot{y}_p + z_p\dot{z}_p) r_p^{-1} \\ (\dot{x}_p r_p - x_p\dot{r}_p) r_p^{-2} \\ (\dot{y}_p r_p - y_p\dot{r}_p) r_p^{-2} \\ (\dot{z}_p r_p - z_p\dot{r}_p) r_p^{-2} \end{bmatrix} \end{aligned}$$
$$\mathbf{\Lambda}_{rr} = \begin{bmatrix} \mathbf{J}_1 \\ \mathbf{J}_2 \end{bmatrix}\mathbf{P}_p\left[\mathbf{J}_1^T\ \mathbf{J}_2^T\right] = \begin{bmatrix} \tau_{rr} & \cdots & \tau_{\dot{r}r} & \tau_{r\dot{\alpha}} & \tau_{r\dot{\beta}} & \tau_{r\dot{\gamma}} \\ \vdots & \ddots & & & & \\ \tau_{\dot{r}r} & & \tau_{\dot{r}\dot{r}} & \tau_{\dot{r}\dot{\alpha}} & \tau_{\dot{r}\dot{\beta}} & \tau_{\dot{r}\dot{\gamma}} \\ \tau_{r\dot{\alpha}} & & \tau_{\dot{r}\dot{\alpha}} & \tau_{\dot{\alpha}\dot{\alpha}} & \tau_{\dot{\alpha}\dot{\beta}} & \tau_{\dot{\alpha}\dot{\gamma}} \\ \tau_{r\dot{\beta}} & & \tau_{\dot{r}\dot{\beta}} & \tau_{\dot{\alpha}\dot{\beta}} & \tau_{\dot{\beta}\dot{\beta}} & \tau_{\dot{\beta}\gamma} \\ \tau_{r\dot{\gamma}} & & \tau_{\dot{r}\dot{\gamma}} & \tau_{\dot{\alpha}\dot{\gamma}} & \tau_{\dot{\beta}\gamma} & \tau_{\dot{\gamma}\dot{\gamma}} \end{bmatrix}^T. \tag{21}$$

The estimate error covariance is computed by the coordinate transformation of the error covariance to direction cosine coordinates. The Jacobian matrix $\mathbf{J}_1$ is given in (11) and $\mathbf{J}_2$ is

$$\mathbf{J}_2 = \left.\frac{\partial \mathbf{h}_2}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_p} = \left[\left.\frac{\partial \dot{r}}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_p} \ \left.\frac{\partial \dot{\alpha}}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_p} \ \left.\frac{\partial \dot{\beta}}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_p} \ \left.\frac{\partial \dot{\gamma}}{\partial \mathbf{x}}\right|_{\hat{\mathbf{x}}_p} \right]^T$$

$$\frac{\partial \dot{r}}{\partial \mathbf{x}} = r^{-2}\begin{bmatrix} \dot{x}r - x\dot{r} \\ \dot{y}r - y\dot{r} \\ \dot{z}r - z\dot{r} \\ xr \\ yr \\ zr \end{bmatrix}$$

$$\frac{\partial \dot{\alpha}}{\partial \mathbf{x}} = r^{-4}\begin{bmatrix} -2x\dot{x}r + (3x^2 - r^2)\dot{r} \\ (-x\dot{y} - y\dot{x})r + 3xy\dot{r} \\ (-x\dot{z} - z\dot{x})r + 3xz\dot{r} \\ (y^2 + z^2)r \\ -xyr \\ -xzr \end{bmatrix}$$

$$
\frac{\partial \dot{\beta}}{\partial \mathbf{x}} = r^{-4} \begin{bmatrix} (-x\dot{y} - y\dot{x})r + 3xy\dot{r} \\ -2y\dot{y}r + (3y^2 - r^2)\dot{r} \\ (-z\dot{y} - z\dot{y})r + 3xy\dot{r} \\ -xyr \\ (x^2 + z^2)r \\ -yzr \end{bmatrix}
$$

$$
\frac{\partial \dot{\gamma}}{\partial \mathbf{x}} = r^{-4} \begin{bmatrix} (-x\dot{z} - z\dot{x})r + 3xz\dot{r} \\ (-y\dot{z} - z\dot{y})r + 3yz\dot{r} \\ -2z\dot{z}r + (3z^2 - r^2)\dot{r} \\ -xzr \\ -yzr \\ (x^2 + y^2)r \end{bmatrix}. \tag{22}
$$

Then, the derivation of the bias and error covariance of pseudo-linearized range rate $\eta_{\dot{r}}$ can be subsequently accomplished as follows:

$$
\begin{aligned}
\mu_{rr} &= E\left[\eta_{\dot{r}} \,\middle|\, \hat{\mathbf{z}}_{rr}\right] \\
&= (-\xi r_p + \dot{r}_p)(\tau_{\alpha\alpha} + \tau_{\beta\beta} + \tau_{\gamma\gamma}) + \psi \\
\psi &= r_p(\tau_{\dot{\alpha}\alpha} + \tau_{\dot{\beta}\beta} + \tau_{\dot{\gamma}\gamma}) + (\dot{\alpha}_p\tau_{r\alpha} + \dot{\beta}_p\tau_{r\beta} + \dot{\gamma}\tau_{r\gamma}) \\
&\quad + \{\alpha_p(\tau_{\dot{r}\alpha} - \xi\tau_{r\alpha}) + \beta_p(\tau_{\dot{r}\beta} - \xi\tau_{r\beta}) \\
&\quad + \gamma_p(\tau_{\dot{r}\gamma} - \xi\tau_{r\gamma})\} \\
\xi &= \rho\sigma_{\dot{r}}\sigma_r^{-1}
\end{aligned} \tag{23}
$$

$$
\begin{aligned}
\mathbf{R}_{rr} &= E\left[(\eta_{\dot{r}} - \mu_{rr})(\eta_{\dot{r}} - \mu_{rr})^T \,\middle|\, \hat{\mathbf{z}}_{rr}\right] \\
&= (1 - \rho^2)\sigma_{\dot{r}}^2 \\
&\quad + (r_p^2 + \tau_{rr})(\dot{\alpha}_p^2\tau_{\alpha\alpha} + \dot{\beta}_p^2\tau_{\beta\beta} + \dot{\gamma}_p^2\tau_{\gamma\gamma}) \\
&\quad + (r_p^2 + \tau_{rr})(\tau_{\dot{\alpha}\dot{\alpha}}\tau_{\alpha\alpha} + \tau_{\dot{\beta}\dot{\beta}}\tau_{\beta\beta} + \tau_{\dot{\gamma}\dot{\gamma}}\tau_{\gamma\gamma}) \\
&\quad + \{(\dot{r}_p^2 + \tau_{\dot{r}\dot{r}}) - 2\xi\dot{r}_p r_p + \xi^2(r_p^2 + \tau_{rr})\} \\
&\qquad \times (\alpha_p^2\tau_{\alpha\alpha} + \beta_p^2\tau_{\beta\beta} + \gamma_p^2\tau_{\gamma\gamma}) \\
&\quad + \{(2\dot{r}_p^2 + 3\tau_{\dot{r}\dot{r}}) - 4\xi\dot{r}_p r_p + \xi^2(2r_p^2 + 3\tau_{rr})\} \\
&\qquad \times (\tau_{\alpha\alpha}^2 + \tau_{\beta\beta}^2 + \tau_{\gamma\gamma}^2) \\
&\quad + 2\{\dot{r}_p r_p - \xi(r_p^2 + \tau_{rr})\} \\
&\qquad \times (\dot{\alpha}_p\alpha_p\tau_{\alpha\alpha} + \dot{\beta}_p\beta_p\tau_{\beta\beta} + \dot{\gamma}_p\gamma_p\tau_{\gamma\gamma}) \\
&\quad + 2(\tau_{\dot{r}\dot{r}} + \xi^2\tau_{rr})(\tau_{\alpha\alpha}\tau_{\beta\beta} + \tau_{\alpha\alpha}\tau_{\gamma\gamma} + \tau_{\beta\beta}\tau_{\gamma\gamma}) \\
&\quad + \xi^2\upsilon_1 - 2\xi\upsilon_2 + \upsilon_3 \\
\xi &= \rho\sigma_{\dot{r}}\sigma_r^{-1}
\end{aligned} \tag{24}
$$

where ψ in (23) and υ_1, υ_2, and υ_3 in (24) corresponds to the bias and covariance created from cross covariances among estimation errors. Appendix B briefly presents how to derive the terms υ_1, υ_2, and υ_3. However, only the equation of υ_1 is given in the Appendix since these terms encompassing cross covariances is pretty tedious while having a limited influence on the tracking performance. Note that the proposed filter would be able to maintain satisfactory performance even ignoring these cross-covariance terms υ_1, υ_2, υ_3, and ψ. When the initial velocity covariance or the position-velocity cross covariance is estimated greater than actual range-rate measurement accuracy, omitting those cross-correlated terms would lead the DDCMKF to achieve faster convergence with comparable accuracy.

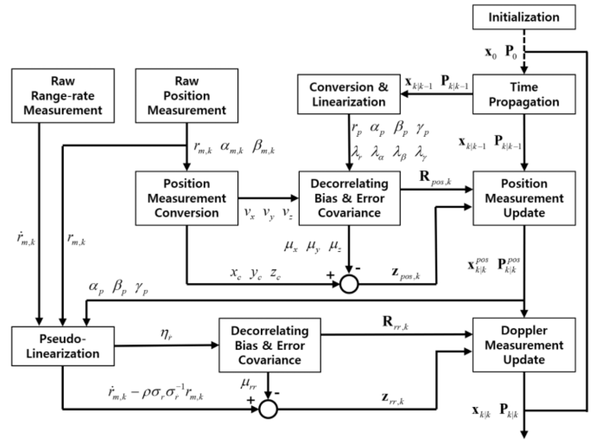


Fig. 2. Proposed DDCMKF algorithm flowchart.

Remark that the proposed method is similar to the sequential pseudo-measurement filter [18] in that it incorporates the range-rate measurement by employing pseudo-linearization. They share two key ideas: the linearization of the range-rate equation using direction cosine estimates and the uncorrelation based on Cholesky factorization. The difference between these two methods originates from the substitution of true direction cosines to estimated direction cosines. In [18], the direction cosine error is linearly approximated in Cartesian coordinates, i.e., the error statistics are assumed to be Gaussian in Cartesian coordinates. The bias and covariance are approximated using the covariance in Cartesian coordinates directly. However, the proposed pseudo-linearization presumes Gaussian estimation errors in the direction cosine coordinate system. It linearly approximates the error covariance. No clear evidence of which approach is better exists because the true error statistics might differ from both assumptions. The proposed pseudo-linearization may provide faster initial convergence at the early steps of the track, as mentioned in the previous section.

C. Filtering Algorithm Formulation

The proposed sequential DDCMKF is formulated in this section. It consists of four substeps as shown in Fig. 2. The filter processes position measurements first, and then the pseudo-linearized range-rate update is accomplished with the latest position estimate. The filter executes the propagation and the two update steps recursively.

1) *Initialization:*

$$
\begin{aligned}
\mathbf{x}_{0|0} &= \left[\mathbf{p}_{c,0} \ \mathbf{v}_{c,0}\right]^T \\
\mathbf{p}_{c,0} &= \left[x_{c,0} \ y_{c,0} \ z_{c,0}\right] \\
\mathbf{v}_{c,0} &= \frac{1}{T}\left[(x_{c,0} - x_{c,-1}) \ (y_{c,0} - y_{c,-1}) \ (z_{c,0} - z_{c,-1})\right] \\
\mathbf{P}_{0|0} &= \begin{bmatrix} \mathbf{R}_{c,0} & \mathbf{R}_{c,0}/T \\ \mathbf{R}_{c,0}/T & (\mathbf{R}_{c,0} + \mathbf{R}_{c,-1})/T \end{bmatrix}
\end{aligned} \tag{25}
$$

where $x_{c,0}, y_{c,0}, z_{c,0}$, and $x_{c,-1}, y_{c,-1}, z_{c,-1}$ are converted positions calculated as in (6) from two initial raw position measurements. Since the estimated state information is unavailable for the initialization, the measurement-conditioned error covariances $\mathbf{R}_{c,0}$ and $\mathbf{R}_{c,-1}$ in [2] and [12] are utilized instead.

2) *Time Propagation:*

$$\begin{aligned}\hat{\mathbf{x}}_{k|k-1} &= \mathbf{F}_{k-1}\hat{\mathbf{x}}_{k-1|k-1}\\ \mathbf{P}_{k|k-1} &= \mathbf{F}_{k-1}\mathbf{P}_{k-1|k-1}\mathbf{F}_{k-1}^T + \mathbf{Q}_{k-1}.\end{aligned} \tag{26}$$

where the state transition matrix of the constant velocity model $\mathbf{F}_{k-1}$ and the process noise covariance $\mathbf{Q}_k$ are given in (3).

3) *Converted Position Measurement Update:*

$$\begin{aligned}\hat{\mathbf{x}}_{pos,k|k} &= \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_{pos,k}(\mathbf{z}_{pos,k} - \mathbf{H}_{pos,k}\hat{\mathbf{x}}_{k|k-1} - \mu_{pos,k})\\ \mathbf{P}_{pos,k|k} &= (\mathbf{I} - \mathbf{K}_{pos,k}\mathbf{H}_{pos,k})\mathbf{P}_{k|k-1}(\mathbf{I} - \mathbf{K}_{pos,k}\mathbf{H}_{pos,k})^T\\ &\quad + \mathbf{K}_{pos,k}\mathbf{R}_{pos,k}\mathbf{K}_{pos,k}^T\\ \mathbf{K}_{pos,k} &= \mathbf{P}_{k|k-1}\mathbf{H}_{pos,k}^T\left[\mathbf{H}_{pos,k}\mathbf{P}_{k|k-1}\mathbf{H}_{pos,k}^T + \mathbf{R}_{pos,k}\right]^{-1}\\ \mathbf{H}_{pos,k} &= \left[\,\mathbf{I}_{3\times3}\ \mathbf{0}_{3\times3}\,\right],\end{aligned} \tag{27}$$

where the converted position measurement $\mathbf{z}_{pos,k}$, its bias $\mu_{pos,k}$, and its error covariance $\mathbf{R}_{pos,k}$ are specified in (6), (12), and (13).

4) *Psuedo-Linearized Range-Rate Measurement Update:*

$$\begin{aligned}\hat{\mathbf{x}}_{k|k} &= \hat{\mathbf{x}}_{pos,k|k} + \mathbf{K}_{rr,k}(\mathbf{z}_{rr,k} - \mathbf{H}_{rr,k}\hat{\mathbf{x}}_{pos,k|k} - \mu_{rr,k})\\ \mathbf{P}_{k|k} &= (\mathbf{I} - \mathbf{K}_{rr,k}\mathbf{H}_{rr,k})\mathbf{P}_{pos,k|k}(\mathbf{I} - \mathbf{K}_{rr,k}\mathbf{H}_{rr,k})^T\\ &\quad + \mathbf{K}_{rr,k}\mathbf{R}_{rr,k}\mathbf{K}_{rr,k}^T\\ \mathbf{K}_{rr,k} &= \mathbf{P}_{pos,k|k}\mathbf{H}_{rr,k}^T\left[\mathbf{H}_{rr,k}\mathbf{P}_{pos,k|k}\mathbf{H}_{rr,k}^T + \mathbf{R}_{rr,k}\right]^{-1}\\ \mathbf{H}_{rr,k} &= \left[-\rho\sigma_r\sigma_{\dot{r}}^{-1}\mathbf{\Delta}_{pos,k|k} \quad \mathbf{\Delta}_{pos,k|k}\right]\\ \mathbf{\Delta}_{pos,k|k} &= \left[\alpha_{pos,k|k} \quad \beta_{pos,k|k} \quad \gamma_{pos,k|k}\right]\\ &= r_{pos,k|k}^{-1}\left[x_{pos,k|k} \quad y_{pos,k|k} \quad z_{pos,k|k}\right],\end{aligned} \tag{28}$$

where the pseudo-linearized range-rate measurement $\mathbf{z}_{rr,k}$, its bias $\mu_{rr,k}$, and its error covariance $\mathbf{R}_{rr,k}$ are presented in (18), (23), and (24), respectively. The covariance update in (27) and (28) adopts the Joseph form to guarantee positive semidefiniteness since lower arithmetic precision under numerical instability might bring about a deadly indefinite covariance matrix. Nevertheless, the simplified form update equation would also be applicable when trying to decrease computation.

Since the range-rate measurement update is handled in sequence using the latest state estimate with little estimation bias, the potential problems of sequential filtering such as estimation accuracy degradation or divergence due to biased estimation are prevented. Note that the two-point difference method [28] is adopted for the initialization. The other tracking initiation algorithms such as the single-point track initiation [29], [30] are also applicable for some cases.

IV. NUMERICAL SIMULATIONS

A. Consistency Evaluation of Converted Measurement Model

The statistical consistency test is required to verify that the measurement conversion used in the filter virtually represents the statistical distribution of the actual converted measurement. The normalized error squared (NES) test [3], [4] is utilized to evaluate the consistency of the model. The statistical consistency of the proposed decorrelated debiased converted measurement (DDCM) model is examined and compared with the conventional debiased converted measurement (CDCM) model used in [2], [12], and [17]. The proposed DDCM includes the pseudo-linearized range-rate measurement, whereas CDCM introduces the pseudo-measurement, the product of the range, and range-rate measurement. The consistency of position-only models is first evaluated to check the effect of the decorrelation used in DDCM. The range-rate measurement model is then augmented to the converted position model for consistency evaluation. Note that the calculation of the DDCM or the range-rate augmented DDCM requires the predicted estimates of the target state. The predicted estimate state vector is assumed to be the true target state vector corrupted by correlated Gaussian noise.

The measurement conversion is regarded to be consistent if the sample mean of NES is located within 1% and 99% probability bounds of the chi-square distribution with $n \times N$ degree of freedom as follows:

$$\begin{aligned}\frac{1}{nN}\chi^2_{n\times N}(0.01) \le \bar{\varphi} \le \frac{1}{nN}\chi^2_{n\times N}(0.99)\\ \bar{\varphi} = \frac{1}{n}\sum_{i=1}^{N}\tilde{\mathbf{z}}_i^T\mathbf{R}_i^{-1}\tilde{\mathbf{z}}_i \sim \frac{1}{nN}\chi^2_{n\times N}\end{aligned} \tag{29}$$

where n denotes the number of components in the measurement error vector $\tilde{\mathbf{z}}$, and N is the number of measurement samples used for the NES test. The measurement error $\tilde{\mathbf{z}}_i$ and measurement covariance $\mathbf{R}_i$ of the DDCM are defined in (30) if the test statistics include both position and range-rate measurements

$$\begin{aligned}\tilde{\mathbf{z}}_i &= \begin{bmatrix}\mathbf{z}_{pos,i}\\ \mathbf{z}_{rr,i}\end{bmatrix} - \begin{bmatrix}\mathbf{I}_{3\times3} & \mathbf{0}_{1\times3}\\ -\rho\sigma_{\dot{r}}\sigma_r^{-1}\mathbf{\Delta}_p & \mathbf{\Delta}_p\end{bmatrix}\mathbf{x} - \begin{bmatrix}\mu_{pos,i}\\ \mu_{rr,i}\end{bmatrix}\\ \mathbf{R}_i &= \begin{bmatrix}\mathbf{R}_{pos,i} & \mathbf{0}_{3\times1}\\ \mathbf{0}_{1\times3} & \mathbf{R}_{rr,i}\end{bmatrix},\end{aligned} \tag{30}$$

where the direction cosine estimate $\mathbf{\Delta}_p$, the converted position and pseudo-linearized range-rate measurements $\mathbf{z}_{pos}$, $\mathbf{z}_{rr}$, and their biases μ_{pos}, μ_{rr} and covariances $\mathbf{R}_{pos}$, $\mathbf{R}_{rr}$ can be found in (12), (13), (23), and (24). For the CDCM, the measurement error and covariance are defined as follows:

$$\begin{aligned}\tilde{\mathbf{z}}_i &= \begin{bmatrix}\mathbf{z}_{pos,i}\\ r_m\dot{r}_m\end{bmatrix} - \begin{bmatrix}\mathbf{x}_{pos}\\ \mathbf{x}_{pos}\cdot\mathbf{x}_{vel}\end{bmatrix} - \begin{bmatrix}\mathbf{u}_{xyz}\\ u_\xi\end{bmatrix}\\ \mathbf{R}_i &= \mathbf{R}_{Car}.\end{aligned} \tag{31}$$

Note that the pseudo-measurement of the CDCM is constructed as the product of the range and range-rate measurement, which is unable to be linearly formulated as

TABLE I
Test Conditions for Consistency Evaluation

Parameters	Case 1 (Long-range Scan)	Case 2 (Short-range Scan)
Range (r, km)	250	50
Azimuth (φ, deg)	0 ~ 360	0 ~ 360
Off-boresight angle (θ, deg)	0 ~ 90	0 ~ 90
Speed (v, m/s)	100	150
Flight path angle (γ, deg)	-90 ~ 90	-90 ~ 90
Course angle (χ, deg)	0 ~ 360	0 ~ 360
Range error SD* (σ_r, m)	100	50
Direction cosine error SD ($\sigma_\alpha, \sigma_\beta$, -)	0.005	0.015
Range-rate error SD ($\sigma_{\dot{r}}$, m/s)	1.0	0.1
Correlation coefficient (ρ, -)	0.0	-0.9
Estimated position error SD (τ_{xyz}, m)	1000	200
Estimated velocity error SD ($\tau_{\dot{x}\dot{y}\dot{z}}$, m/s)	40	25
Correlation between estimated states (ρ, -)	0.3	0.3

(*) SD: Standard deviation.

in (30). Thus, the pseudo-measurement is defined as the dot product of the position and velocity where $\mathbf{x}_{pos} = [x\ y\ z]^T$ and $\mathbf{x}_{vel} = [\dot{x}\ \dot{y}\ \dot{z}]^T$. The equation for bias components $\mathbf{u}_{xyz}$, u_ξ and the corresponding covariance $\mathbf{R}_{Car}$ can be found in [17].

Two cases of NES tests are performed. In the first case, long-range targets with relatively small direction cosine measurement errors are assumed. The other case considers short-range targets. It is notable that the second case, unlike the first case, assumes a relatively large direction cosine measurement error. The negative correlation between range error and range-rate error, expected during practical tracking as noted in [14], is also considered. The details of the parameters used in NES tests are stated in Table I. As indicated in Fig. 1, the z-axis corresponds to the boresight of a radar, which means xy-plane is a radar face plane with an off-boresight angle $\theta = 90°$. The NES tests are performed for every azimuth and the off-boresight angle at 1° interval, respectively. Each test considers 10^4 samples of constant velocity targets. The flight path and course of targets are random in all directions. A consistent measurement model is expected to yield an NES value equal to unity.

Fig. 3 shows the average NES plots of the DDCM and the CDCM for the first case, scanning long-range targets. Both the DDCM and the CDCM yield consistent NES for all directions over the radar face plane, except when the off-boresight angle is high. The DDCM provides consistency within the off-boresight angle 76° for both cases considering only position measurements and incorporating range-rate measurements. The covariance for the points whose off-boresight angle is higher than 76° is overestimated in the DDCM; therefore, the sample mean of NES is below the lower bound of consistency as shown in Fig. 3(a) and (c). The CDCM underestimates the covariance when the off-boresight angle is higher than 55° as shown in Fig. 3(b) and (d).

Fig. 4 shows the average NES of the DDCM and the CDCM for the second case. The noisy angular measurement leads to dramatic performance degradation of the CDCM. The significant bias originating from measurement-conditioned covariance makes the CDCM provide consistency only for the area within off-boresight angle smaller than 15° even for the case considering only position measurements. However, the DDCM model suggested in this study grants consistency within the off-boresight angle 67° for the case considering position measurements and 62° for the case embodying range-rate measurements. The adoption of the DDCM in the DDCMKF algorithm can improve the filter accuracy while extending the field of tracking. The DDCM model shows an expanded consistent boundary compared with the CDCM model.

B. Targets Tracking Simulation Results

The performance of the proposed tracking filter is validated through numerical simulations and performance comparisons with other filtering algorithms in this section. The DDCMKF is compared against the following seven types of filtering that consider measurements in the direction cosine coordinate system:

1) the DCMSEKF incorporating converted position and range-rate measurements [17];
2) the EKF with raw measurements in direction cosine coordinates [15];
3) the UKF with raw measurements [31];
4) the quadrature Kalman filter (QKF) considering raw measurements [32];
5) the cubature Kalman filter (CKF) processing raw measurements [33];
6) the PCMKF considering only converted position measurements [2], [12];
7) the recursive LMMSE filter in direction cosine coordinates [27].

Note that all algorithms except two algorithms (6) and (7) consider range-rate measurements. The UKF employs $\alpha_{UT} = 10^{-3}$, $\beta_{UT} = 2$, and $\kappa_{UT} = 1$ in the unscented transformation. The QKF sets three-quadrature points for each state variable, which is 729 points in total.

Two cases of Monte-Carlo simulations are implemented to verify the performance of the DDCMKF. In every simulation, the radar geometry is identical to that used in the consistency test. The radar measures the range, two-direction cosines to face plane, and range rate of targets with 1-Hz frequency. Totally 10^4 constant velocity targets are generated with random initial position and velocity on the virtual hemisphere over the radar face plane. The detailed simulation conditions and the parameters for noise or uncertainty are given in Table II. Two Monte-Carlo simulation scenarios are identical to scenarios used in the consistency evaluation except that the initial off-boresight angle of targets is limited within 60° referring to consistency

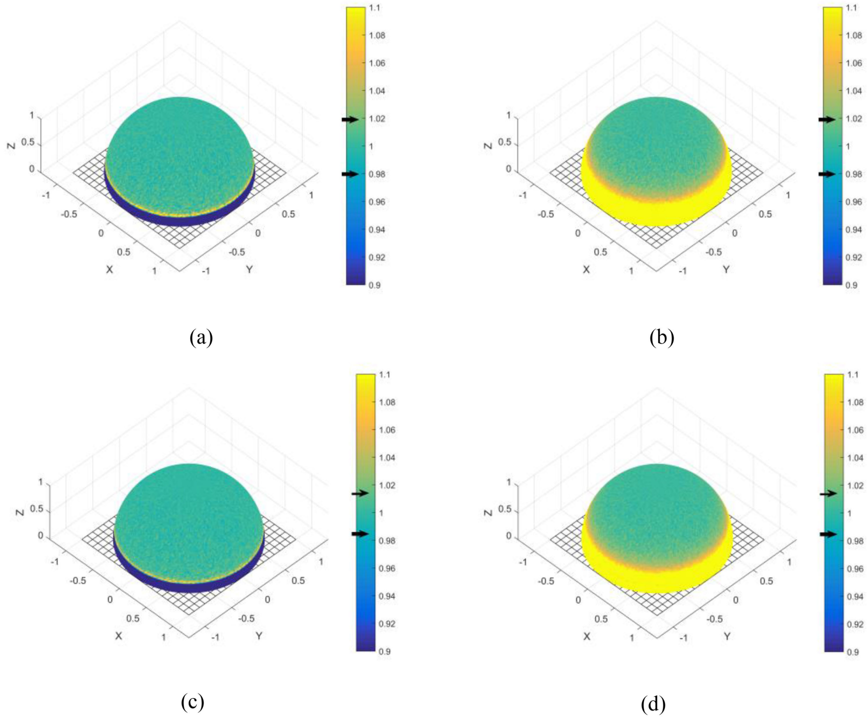


Fig. 3. Average NES plots for the first scenario of the DDCM model and the CDCM model (plotting on unit hemispheres above radar face plane, marking chi-square probability bounds (0.99, 0.01) on color bars with cursors). (a) DDCM NES (Position only). (b) CDCM NES (Position only). (c) DDCM NES (Incl. range-rate). (d) CDCM NES (Incl. range-rate).

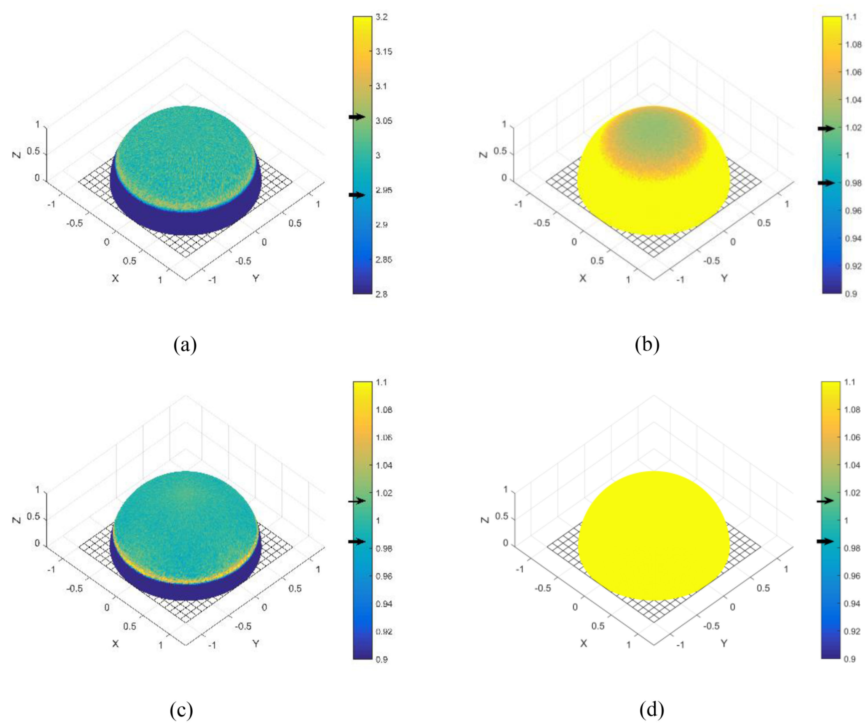
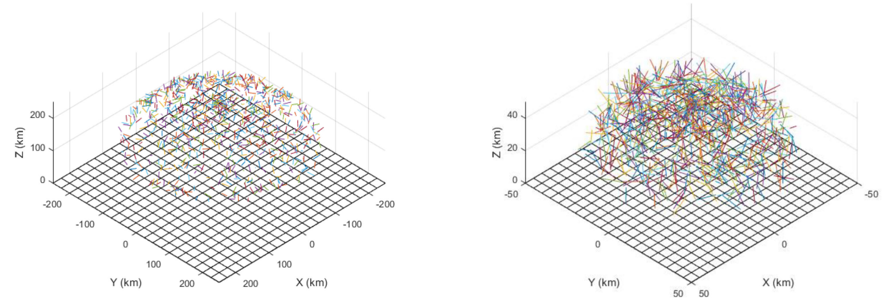


Fig. 4. Average NES plots for the second scenario of the DDCM model and the CDCM model (plotting on unit hemispheres above radar face plane, marking chi-square probability bounds (0.99, 0.01) on color bars with cursors). (a) DDCM NES (Position only). (b) CDCM NES (Position only). (c) DDCM NES (Incl. range-rate). (d) CDCM NES (Incl. range-rate).



(a)

(b)

Fig. 5. Illustration of targets used in Monte-Carlo simulations (showing targets' trajectories). (a) Long-range targets (Scenario 1). (b) Short-range targets (Scenario 2).

TABLE II
Scenario Parameters for Monte-Carlo Simulations

Parameters	Scenario 1 (Long-range Scan)	Scenario 2 (Short-range Scan)
Initial range (r, km)	250	50
Initial azimuth (φ, deg)	0 ~ 360	0 ~ 360
Initial off-boresight angle (θ, deg)	0 ~ 60	0 ~ 60
Initial speed (v, m/s)	100	150
Initial flight path angle (γ, deg)	-90 ~ 90	-90 ~ 90
Initial course angle (χ, deg)	0 ~ 360	0 ~ 360
Range error SD* (σ_r, m)	100	50
Direction cosine error SD ($\sigma_\alpha, \sigma_\beta$, -)	0.005	0.015
Range-rate error SD ($\sigma_{\dot{r}}$, m/s)	1.0	0.1
Correlation coefficient (ρ, -)	0.0	-0.9
Process noise PSD (q_{xyz}, m^2/s^3)	0.007	0.0016

(∗) SD: Standard deviation, PSD: Power spectral density.

test results. This limitation is also widely applied to prevent radar from performance degradation in practical tracking [38]. Simulations are performed for 100 s. Targets generated according to two scenarios are depicted in Fig. 5(a) and (b). Two metrics, the root mean squared errors (RMSE) and the average normalized estimation error squared (ANEES) [34] are utilized to verify the performance of the proposed filtering algorithm. Equations for calculating the position and velocity RMSE and the ANEES are given as follows:

$$\mathrm{RMSE}_k^{pos} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left[(\hat{x}_{k|k}^i - x_k^i)^2 + (\hat{y}_{k|k}^i - y_k^i)^2 + (\hat{z}_{k|k}^i - z_k^i)^2\right]}$$
$$\mathrm{RMSE}_k^{vel} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left[(\hat{\dot{x}}_{k|k}^i - \dot{x}_k^i)^2 + (\hat{\dot{y}}_{k|k}^i - \dot{y}_k^i)^2 + (\hat{\dot{z}}_{k|k}^i - \dot{z}_k^i)^2\right]}$$
$$\mathrm{ANEES} = \frac{1}{nN}\sum_{i=1}^{N}(\hat{\mathbf{x}}_{k|k}^i - \mathbf{x}_{k|k}^i)^T \mathbf{P}_{k|k}^i (\hat{\mathbf{x}}_{k|k}^i - \mathbf{x}_{k|k}^i), \quad (32)$$

where $\hat{\mathbf{x}}_k^i = [\hat{x}_k^i, \hat{y}_k^i, \hat{z}_k^i, \hat{\dot{x}}_k^i, \hat{\dot{y}}_k^i, \hat{\dot{z}}_k^i]^T$ and $\mathbf{x}_k^i = [x_k^i, y_k^i, z_k^i, \dot{x}_k^i, \dot{y}_k^i, \dot{z}_k^i]^T$ represent the estimated and actual target state vector at the time step k of simulation number i. The number of simulation and state variables are denoted as N and n each. The position and velocity RMSE show the convergence speed and tracking performance of filters, whereas the ANEES examines the consistency of filters' estimation. The posterior Cramer–Rao lower bound (PCRLB) [35]–[37] that demonstrates the lower limit of the estimator variance is also computed to measure the performance of filters. The position and velocity PCRLB can be obtained from taking the inverse of the information matrix $\mathbf{J}_k$, which is updated as follows:

$$\mathbf{J}_k = (\mathbf{Q}_{k-1} + \mathbf{F}_{k-1}\mathbf{J}_{k-1}^{-1}\mathbf{F}_{k-1}^T)^{-1} + E\left[\mathbf{H}_k^T \mathbf{R}_k^{-1}\mathbf{H}_k\right]$$
$$\mathbf{H}_k = \left[\left.\frac{\partial r}{\partial \mathbf{x}}\right|_{\mathbf{x}} \left.\frac{\partial \alpha}{\partial \mathbf{x}}\right|_{\mathbf{x}} \left.\frac{\partial \beta}{\partial \mathbf{x}}\right|_{\mathbf{x}} \left.\frac{\partial \dot{r}}{\partial \mathbf{x}}\right|_{\mathbf{x}}\right]^T$$
$$\mathbf{J}_0 = \mathbf{P}_{0|0}^{-1} \quad (33)$$

where derivatives in Jacobian matrix $\mathbf{H}_k$ can be found in (11) and (21). Since the second term on the information matrix update equation nonlinearly depends on the actual random state, it cannot be estimated analytically. Accordingly, it is closely approximated in a Monte-Carlo manner by taking an average of a large number of samples. The expectation notation can be consequently omitted, and then the information matrix is recursively estimated. The position and velocity PCRLB, the inverse the information matrix, are demonstrated on the RMSE plots as a reference for the lower estimation bound as follows:

$$\mathrm{CRLB}\{[\hat{\mathbf{x}}_k^i]_j\} = [\mathbf{J}_k^{-1}]_{jj}^i \quad (i = 1, \cdots, 6)$$
$$\mathrm{CRLB}_{pos} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{3}[\mathbf{J}_k^{-1}]_{jj}^i}$$
$$\mathrm{CRLB}_{vel} = \sqrt{\sum_{i=1}^{N}\sum_{j=4}^{6}[\mathbf{J}_k^{-1}]_{jj}^i}. \quad (34)$$

Furthermore, the computation complexity of filters is examined by mean execution CPU time spent during the Monte-Carlo simulations. It is not a significant concern for the filters' performance, yet the filter with the less computational requirement has a comparative advantage in

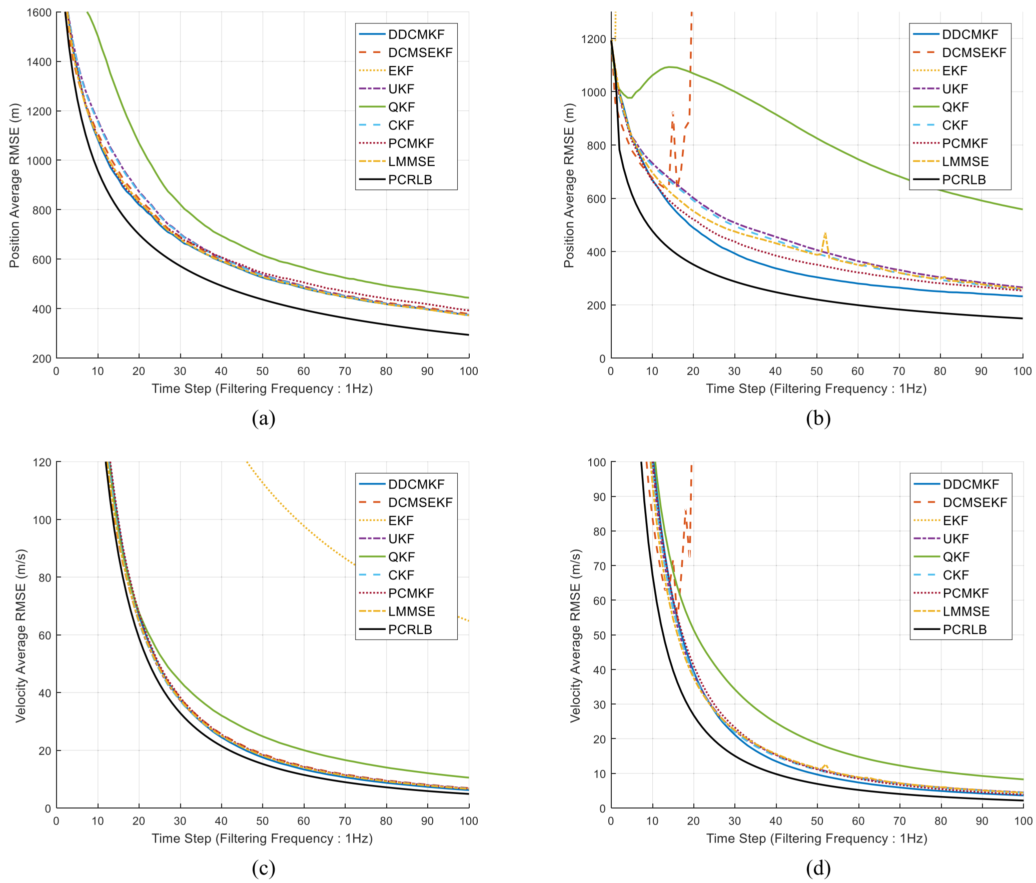


Fig. 6. Position and velocity error performance comparison for 10^4 runs of Monte-Carlo simulations (the position RMSE of the EKF in scenario 1 increases up to 12 000 m and is neglected from the plot for comparison of other algorithms). (a) Long-range tracking position RMSE (Scenario 1). (b) Short-range tracking position RMSE (Scenario 2). (c) Long-range tracking velocity RMSE (Scenario 1). (d) Short-range tracking velocity RMSE (Scenario 2).

that it can reduce the computational burden of an on-board flight radar computer.

The error comparison of tracking algorithms in the long-range tracking scenario shows that the proposed DDCMKF slightly outperforms other tracking filters by showing more minor errors. From the position and velocity RMSE in Fig. 6(a) and (c), the proposed DDCMKF exhibits more than 30 m smaller position error after 25 s and 1 m/s smaller velocity error after 50 s compared to other algorithms. On account of linearization on noisy position measurements, the performance of EKF is the poorest among all other algorithms.

The DDCMKF surpasses other algorithms in tracking short-range targets under highly noisy angular measurement as shown in Fig. 6(b) and (d). Since scenario 2 assumes significant angular measurement error and the correlation between the range and range-rate error, it clearly delineates the effectiveness of the decorrelation used in the DDCMKF. The proposed DDCMKF is robust to noisy position measurements while the large estimation bias originated from position estimation leads the DCMSEKF to diverge. The proposed DDCMKF demonstrates relatively faster convergence than any other methods after 10 s as in Fig. 6(b), whereas its convergence during the initial 10 s is marginally slow. The proposed DDCMKF demonstrates about 100 m smaller position RMSE than the UKF, the CKF, and the QKF after 40 s of tracking. It also achieves about 1 m/s smaller velocity RMSE than other algorithms after 35 s as well. It seems that the DDCMKF shows a slightly slower or comparable convergence speed during the early steps of tracking compared to others. The initial position measurements with substantial error as in scenario 2 lead the two-point initialization to overrate the initial velocity cross-covariance terms in (13) and (24), which prevents the DDCMKF from converging fast. Neglecting cross-covariance terms, as described below (24), would help achieve faster convergence. In any simulation case, the proposed DDCMKF shows excellent position tracking performance undoubtedly.

The ANEES analysis results of Fig. 7(a) and (b) reveal the consistency of the proposed DDCMKF. Although all six filtering algorithms employing the range-rate measurement initially deviate from the credibility bounds, the ANEES of the DDCMKF rapidly converges and remains close to the consistent boundary during simulations. Although the ANEES of the DDCMKF slightly strays from the credibility

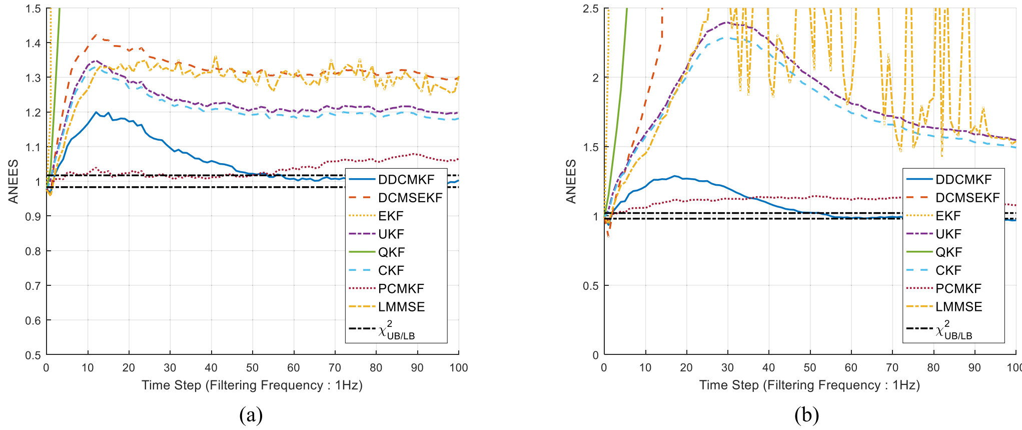


Fig. 7. Consistency test results of filtering algorithms for 10^4 runs of Monte-Carlo simulations (the ANEES of the EKF and the QKF in both scenarios drastically diverges from the initiation of tracking and is removed from the plot). (a) ANEES for long-range tracking (Scenario 1). (b) ANEES for short-range tracking (Scenario 2).

TABLE III
Mean CPU Time Per Run in Monte-Carlo Simulations

Filter Algorithms	Mean CPU Time (ms/run)
DDCMKF	0.169
DCMSEKF	0.161
EKF	0.090
UKF	0.481
QKF	9.10
CKF	0.301
PCMKF	0.147
LMMSE	0.263

boundary in scenario 2, it would be the most credible filter among these algorithms. The ANEES of the DCMSEKF, the EKF, the UKF, the QKF, the CKF, and the LMMSE show that their estimation results remain optimistic during the whole simulation. A highly nonlinear condition of scenario 2 brings about slower convergence of the ANEES of the DDCMKF. The PCMKF that does not utilize nonlinear range-rate measurements remains close to the ANEES credibility region for both scenarios; however, it slowly diverges due to an estimation bias.

The mean CPU times spent per a single run of filters in Monte-Carlo simulations are given in Table III. The proposed DDCMKF requires 0.169 ms CPU time for a single run on average. The mean CPU time of the DDCMKF is of a similar magnitude to the DCMSEKF. Though covariance equations of the DDCMKF are rather complicated than the DCMSEKF, it is comparable with the DCMSEKF since the algorithm is computationally simple with fewer matrix inversions. The EKF requires the lowest computational load; however, its tracking performance is the poorest among all. The UKF, the CKF, and the LMMSE require a longer computation time than the DDCMKF for a single run due to transformations included in algorithms. The massive computation load of the QKF is not desirable at all albeit their robust performance. In conclusion, the proposed DDCMKF improves the tracking performances while demanding a similar level of computation load to others.

V. CONCLUSION

In this article, the target tracking problem with position and range-rate measurements specified in direction cosine coordinates is considered. A new tracking filter called the DDCMKF is proposed to alleviate the estimation bias commonly exhibited in the conventional CMKF by the measurement-dependent bias and error covariance. The decorrelated position measurement conversion and the pseudo-linearization of the range-rate measurement are simply introduced by replacing true states with estimated states and corresponding estimation errors. The decorrelation of converted measurement error can enlarge the consistent boundary as well as enhance the tracking performance. The adoption of decorrelation further complements the stability of sequential filtering in the range-rate measurement update. Comparative Monte-Carlo simulations verify the tracking performance of the proposed DDCMKF with other nonlinear filtering algorithms. The proposed DDCMKF shows superior performance and more consistent estimation than the other algorithms while requiring relatively low computation loads.

APPENDIX

A. Derivation of Expectations $E[v_\gamma|\alpha_p, \beta_p]$, $E[v_\alpha v_\gamma|\alpha_p, \beta_p]$, $E[v_\beta v_\gamma|\alpha_p, \beta_p]$ and $E[v_\gamma^2|\alpha_p, \beta_p]$

The Taylor expansion of the third direction cosine error v_γ is given in (7). The partial derivatives in (7) are specified as follows:

$$\gamma_\alpha = -\frac{\alpha}{\gamma}, \quad \gamma_\beta = -\frac{\beta}{\gamma}$$

$$\gamma_{\alpha a} = -\frac{1-\beta^2}{\gamma^3}, \quad \gamma_{\alpha\beta} = -\frac{\alpha\beta}{\gamma^3}, \quad \gamma_{\beta\beta} = -\frac{1-\alpha^2}{\gamma^3}$$

$$\gamma_{\alpha\alpha\alpha} = -\frac{3\alpha(1-\beta^2)}{\gamma^5}, \quad \gamma_{\alpha\alpha\beta} = -\frac{\beta(3\alpha^2+\gamma^2)}{\gamma^5}$$
$$\gamma_{\alpha\beta\beta} = -\frac{\alpha(3\beta^2+\gamma^2)}{\gamma^5}, \quad \gamma_{\beta\beta\beta} = -\frac{3\beta(1-\alpha^2)}{\gamma^5}$$
$$\gamma_{\alpha\alpha\alpha\alpha} = -\frac{3(1-\beta^2)(5\alpha^2+\gamma^2)}{\gamma^7},$$
$$\gamma_{\alpha\alpha\alpha\beta} = -\frac{3\alpha\beta(5\alpha^2+3\gamma^2)}{\gamma^7}$$
$$\gamma_{\alpha\alpha\beta\beta} = \frac{15\alpha^2\beta^2+\gamma^2(3-2\gamma^2)}{\gamma^7}$$
$$\gamma_{\alpha\beta\beta\beta} = -\frac{3\alpha\beta(5\beta^2+3\gamma^2)}{\gamma^7}$$
$$\gamma_{\beta\beta\beta\beta} = -\frac{3(1-\alpha^2)(5\beta^2+\gamma^2)}{\gamma^7}. \tag{35}$$

The expectation of third-direction cosine error $E[\upsilon_\gamma|\alpha_p, \beta_p]$ conditioned on two estimated direction cosines (α_p, β_p) is derived using (35). The nested conditioning method [2]–[4], [12], [13], [17] is utilized in the derivation. The direct conditioning suggested in [5], [6], and [8] is also applicable, yet the expression becomes extremely complicated while the difference in resultant equations is negligible. The nested conditioning begins with taking the expectation of direction cosine error with respect to true states as follows:

$$E\left[\upsilon_\gamma\middle|\alpha,\beta\right] = \frac{1}{2}\gamma_{\alpha\alpha}\sigma_\alpha^2 + \frac{1}{2}\gamma_{\beta\beta}\sigma_\beta^2 + \frac{1}{8}\gamma_{\alpha\alpha\alpha\alpha}\sigma_\alpha^4 + \frac{1}{4}\gamma_{\alpha\alpha\beta\beta}\sigma_\alpha^2\sigma_\beta^2 + \frac{1}{8}\gamma_{\beta\beta\beta\beta}\sigma_\beta^4. \tag{36}$$

Then, conditioning (36) with the predicted estimate yields the expectation of the third direction cosine error as follows:

$$\begin{aligned} E\left[\upsilon_\gamma\middle|\alpha_p,\beta_p\right] &= \frac{1}{2}E\left[\gamma_{\alpha\alpha}\middle|\alpha_p,\beta_p\right]\sigma_\alpha^2 \\ &+ \frac{1}{2}E\left[\gamma_{\beta\beta}\middle|\alpha_p,\beta_p\right]\sigma_\beta^2 \\ &+ \frac{1}{8}E\left[\gamma_{\alpha\alpha\alpha\alpha}\middle|\alpha_p,\beta_p\right]\sigma_\alpha^4 \\ &+ \frac{1}{8}E\left[\gamma_{\beta\beta\beta\beta}\middle|\alpha_p,\beta_p\right]\sigma_\beta^4 \\ &+ \frac{1}{4}E\left[\gamma_{\alpha\alpha\beta\beta}\middle|\alpha_p,\beta_p\right]\sigma_\alpha^2\sigma_\beta^2, \end{aligned} \tag{37}$$

where the expectations of derivatives $E[\gamma_{\alpha\alpha}|\alpha_p, \beta_p]$ and $E[\gamma_{\alpha\alpha\alpha\alpha}|\alpha_p, \beta_p]$ are approximately calculated as follows:

$$\begin{aligned} E\left[\gamma_{\alpha\alpha}\middle|\alpha_p,\beta_p\right] &\simeq -\frac{E\left[1-\beta^2\middle|\alpha_p,\beta_p\right]}{E\left[\gamma^2\middle|\alpha_p,\beta_p\right]^{(3/2)}} \\ &= -\left\{1-(\beta_p^2+\tau_{\beta\beta})\right\}\left(1-\alpha_p^2-\beta_p^2-\tau_{\alpha\alpha}-\tau_{\beta\beta}\right)^{-3/2} \\ E\left[\gamma_{\alpha\alpha\alpha\alpha}\middle|\alpha_p,\beta_p\right] &\simeq -\frac{E\left[3(1-\beta^2)(5\alpha^2+\gamma^2)\middle|\alpha_p,\beta_p\right]}{E\left[\gamma^2\middle|\alpha_p,\beta_p\right]^{(7/2)}} \\ &= -3\left\{1+4(\alpha_p^2+\tau_{\alpha\alpha})-2(\beta_p^2+\tau_{\beta\beta}) \right. \\ &\quad - 4\left(\alpha_p^2+\tau_{\alpha\alpha}\right)(\beta_p^2+\tau_{\beta\beta}) \\ &\quad \left. + (\beta_p^4+6\beta_p^2\tau_{\beta\beta}+3\tau_{\beta\beta}^2)\right\}(1-\alpha_p^2-\beta_p^2-\tau_{\alpha\alpha}-\tau_{\beta\beta})^{-7/2}. \end{aligned} \tag{38}$$

In the derivation of (38), higher-order moments of Gaussian variables can be obtained by applying Isserlis' theorem. Let $(\lambda_1, \cdots, \lambda_4)$ follows a zero-mean multivariate Gaussian distribution, then the fourth-order moment would be specified as follows:

$$E\left[\lambda_1\lambda_2\lambda_3\lambda_4\right] = E\left[\lambda_1\lambda_2\right]E\left[\lambda_3\lambda_4\right] + E\left[\lambda_1\lambda_3\right]E\left[\lambda_2\lambda_4\right] + E\left[\lambda_1\lambda_4\right]E\left[\lambda_2\lambda_3\right]. \tag{39}$$

In the case of a fourth-order moment, we have $E[\lambda_\alpha] = 3\tau_{\alpha\alpha}^2$ as in (38). The other expectations of derivatives can be computed in the same manner. The remaining expectation of the derivatives $E[\gamma_{\beta\beta}|\alpha_p, \beta_p]$, $E[\gamma_{\alpha\alpha\beta\beta}|\alpha_p, \beta_p]$, and $E[\gamma_{\beta\beta\beta\beta}|\alpha_p, \beta_p]$ can be derived further. The details of the nested conditioning and derivation of these derivatives can be found in [2] and [17].

Note that denominators in (38) have a term $(1-\alpha_p^2-\beta_p^2-\tau_\alpha^2-\tau_\beta^2)^{1/2}$, which might seem to be negative in some cases and bring out unfavorable results. However, this would not happen in a real tracking problem since the DDCMKF adopts the two-point initialization. It would yield predictions (α_p, β_p) close to real measurements and predicted estimate error covariances $\tau_{\alpha\alpha}$, $\tau_{\beta\beta}$ that are bounded by measurement error covariances. In addition, a practical tracking system usually limits operational off-boresight angle [37] for avoiding performance degradation, which means the range of (α_p, β_p) is saturated and the estimated error of direction cosines would also be bounded.

B. Cross-Correlated Terms in the Pseudo-Linearized Range-Rate Covariance

The cross-covariance terms included in the error covariance of the pseudo-linearized range rate (24) are omitted. The derivation of equations υ_1, υ_2, and υ_3 is lengthy and laborious while its significance turns out to be comparably small in practical cases. In this section, the equation of term υ_1 is derived as an example. The other two terms υ_2 and υ_3 can be derived in the similar manner. Isserlis' theorem introduced in Appendix A is used to calculate higher-order cross moments in the pseudo-linearized range-rate covariance. From (20), calculating the error covariance related to the decorrelated position yields the following:

$$\begin{aligned} &E\left[\xi^2(\lambda_\alpha x_t+\lambda_\beta y_t+\lambda_\gamma z_t)^2\middle|\hat{\mathbf{z}}_{rr}\right] \\ &- E\left[\xi(\lambda_\alpha x_t+\lambda_\beta y_t+\lambda_\gamma z_t)\middle|\hat{\mathbf{z}}_{rr}\right]^2 \\ &= E\left[\xi^2\{\lambda_\alpha(r_p-\lambda_r)(\alpha_p-\lambda_\alpha)+\lambda_\beta(r_p-\lambda_r)(\beta_p-\lambda_\beta) \right. \\ &\left. +\lambda_\gamma(r_p-\lambda_r)(\gamma_p-\lambda_\gamma)\}^2\middle|\hat{\mathbf{z}}_{rr}\right] \\ &- \xi^2\{-r_p(\tau_{\alpha\alpha}+\tau_{\beta\beta}+\tau_{\gamma\gamma})-(\alpha_p\tau_{r\alpha}+\beta_p\tau_{r\beta}+\gamma_p\tau_{r\gamma})\}^2 \end{aligned}$$

$$
\begin{aligned}
&= \xi^2\{(r_p^2+\tau_{rr})(\alpha_p^2\tau_{\alpha\alpha}+\beta_p^2\tau_{\beta\beta}+\gamma_p^2\tau_{\gamma\gamma}) \\
&+(2r_p^2+3\tau_{rr})(\tau_{\alpha\alpha}^2+\tau_{\beta\beta}^2+\tau_{\gamma\gamma}^2) \\
&+2\tau_{rr}(\tau_{\alpha\alpha}\tau_{\beta\beta}+\tau_{\alpha\alpha}\tau_{\gamma\gamma}+\tau_{\beta\beta}\tau_{\gamma\gamma})+\upsilon_1\} \qquad (40)
\end{aligned}
$$

where υ_1 containing numerous cross-covariance terms is given as follows:

$$
\begin{aligned}
\upsilon_1 = {} & 10r_p(\alpha_p\tau_{r\alpha}\tau_{\alpha\alpha}+\beta_p\tau_{r\beta}\tau_{\beta\beta}+\gamma_p\tau_{r\gamma}\tau_{\gamma\gamma}) \\
&+2(r_p^2+\tau_{rr})(\alpha_p\beta_p\tau_{\alpha\beta}+\alpha_p\gamma_p\tau_{\alpha\gamma}+\beta_p\gamma_p\tau_{\beta\gamma}) \\
&+4(r_p^2+\tau_{rr})(\tau_{\alpha\beta}^2+\tau_{\alpha\gamma}^2+\tau_{\beta\gamma}^2) \\
&+2(r_p\alpha_p+\tau_{r\alpha})(\tau_{r\alpha}\tau_{\beta\beta}+\tau_{r\alpha}\tau_{\gamma\gamma}+4\tau_{r\beta}\tau_{\alpha\beta}+4\tau_{r\gamma}\tau_{\alpha\gamma}) \\
&+2(r_p\beta_p+\tau_{r\beta})(\tau_{r\beta}\tau_{\alpha\alpha}+\tau_{r\beta}\tau_{\gamma\gamma}+4\tau_{r\alpha}\tau_{\alpha\beta}+4\tau_{r\gamma}\tau_{\beta\gamma}) \\
&+2(r_p\gamma_p+\tau_{r\gamma})(\tau_{r\gamma}\tau_{\alpha\alpha}+\tau_{r\gamma}\tau_{\beta\beta}+4\tau_{r\alpha}\tau_{\alpha\gamma}+4\tau_{r\beta}\tau_{\beta\gamma}) \\
&+(\alpha_p^2\tau_{r\alpha}^2+\beta_p^2\tau_{r\beta}^2+\gamma_p^2\tau_{r\gamma}^2 \\
&+\cdots+2\alpha_p\beta_p\tau_{r\alpha}\tau_{r\beta}+2\alpha_p\gamma_p\tau_{r\alpha}\tau_{r\gamma}+2\beta_p\gamma_p\tau_{r\beta}\tau_{r\gamma}) \\
&+2\tau_{r\alpha}^2(6\tau_{\alpha\alpha}+\tau_{\beta\beta}+\tau_{\gamma\gamma})+2\tau_{r\beta}^2(6\tau_{\beta\beta}+\tau_{\alpha\alpha}+\tau_{\gamma\gamma}) \\
&+2\tau_{r\gamma}^2(6\tau_{\gamma\gamma}+\tau_{\alpha\alpha}+\tau_{\beta\beta}). \qquad (41)
\end{aligned}
$$

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